GPU Computing with OpenACC Directives

A Very Simple Exercise: SAXPY

SAXPY in C

```
void saxpy(int n,
           float a,
           float *x,
           float *restrict y)
{
#pragma acc kernels
  for (int i = 0; i < n; ++i)
    y[i] = a*x[i] + y[i];
}

...
// Perform SAXPY on 1M elements
saxpy(1<<20, 2.0, x, y);
...
```

SAXPY in Fortran

```
subroutine saxpy(n, a, x, y)
  real :: x(:), y(:), a
  integer :: n, i
$!acc kernels
  do i=1,n
    y(i) = a*x(i)+y(i)
  enddo
$!acc end kernels
end subroutine saxpy

...
$ Perform SAXPY on 1M elements
call saxpy(2**20, 2.0, x_d, y_d)
...
```

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Directive Syntax

- Fortran

  `!$acc directive [clause [,] clause] …]`

  Often paired with a matching end directive surrounding a structured code block

  `!$acc end directive`

- C

  `#pragma acc directive [clause [,] clause] …]`

  Often followed by a structured code block

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kernels: Your first OpenACC Directive

Each loop executed as a separate *kernel* on the GPU.

```
!$acc kernels
     do i=1,n
        a(i) = 0.0
        b(i) = 1.0
        c(i) = 2.0
     end do

     do i=1,n
        a(i) = b(i) + c(i)
     end do
!$acc end kernels
```

kernel 1

kernel 2

Kernel:
A parallel function that runs on the GPU

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Kernels Construct

Fortran

```
!$acc kernels [clause …]
    structured block
!$acc end kernels
```

C

```
#pragma acc kernels [clause …]
    { structured block }
```

Clauses

if(condition)

async(expression)

Also, any data clause (more later)

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C tip: the restrict keyword

- Declaration of intent given by the programmer to the compiler

  Applied to a pointer, e.g.

  float *restrict ptr

  Meaning: "for the lifetime of ptr, only it or a value directly derived from it (such as `ptr + 1`) will be used to access the object to which it points"*

- Limits the effects of pointer aliasing
- OpenACC compilers often require **`restrict`** to determine independence
  - Otherwise the compiler can't parallelize loops that access ptr
  - Note: if programmer violates the declaration, behavior is undefined

http://en.wikipedia.org/wiki/Restrict

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Complete SAXPY example code

- Trivial first example
  - Apply a loop directive
  - Learn compiler commands

```
#include <stdlib.h>

void saxpy(int n,
           float a,
           float *x,
           float *restrict y)
{
#pragma acc kernels
  for (int i = 0; i < n; ++i)
    y[i] = a * x[i] + y[i];
}
```

*restrict:
"I promise y does not alias x"

```
int main(int argc, char **argv)
{
  int N = 1<<20; // 1 million floats

  if (argc > 1)
    N = atoi(argv[1]);

  float *x = (float*)malloc(N * sizeof(float));
  float *y = (float*)malloc(N * sizeof(float));

  for (int i = 0; i < N; ++i)

    x[i] = 2.0f;
    y[i] = 1.0f;
  }

  saxpy(N, 3.0f, x, y);

  return 0;
}
```

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Compile and run

- C:

```
pgcc -acc -ta=nvidia -Minfo=accel -o saxpy_acc saxpy.c
```

- Fortran:

```
pgf90 -acc -ta=nvidia -Minfo=accel -o saxpy_acc saxpy.f90
```

- Compiler output:

```
pgcc -acc -Minfo=accel -ta=nvidia -o saxpy_acc saxpy.c
saxpy:
      8, Generating copyin(x[:n-1])
         Generating copy(y[:n-1])
         Generating compute capability 1.0 binary
         Generating compute capability 2.0 binary
      9, Loop is parallelizable
         Accelerator kernel generated
          9, #pragma acc loop worker, vector(256) /* blockIdx.x threadIdx.x */
             CC 1.0 : 4 registers; 52 shared, 4 constant, 0 local memory bytes; 100% occupancy
             CC 2.0 : 8 registers; 4 shared, 64 constant, 0 local memory bytes; 100% occupancy
```

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GPU startup overhead

- If no other GPU process running, GPU driver may be swapped out
  - Linux specific
  - Starting it up can take 1-2 seconds
- Two options
  - Run nvidia-smi in persistence mode (requires root permissions)
  - Run “nvidia-smi –q –l 30” in the background
- If your running time is off by ~2 seconds from results in these slides, suspect this
  - Nvidia-smi should be running in persistent mode for these exercises

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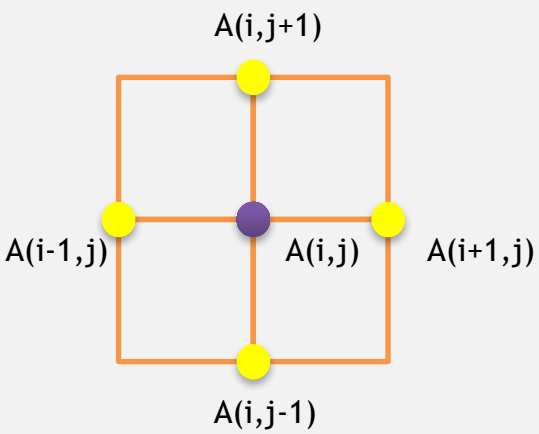
Example: Jacobi Iteration

- Iteratively converges to correct value (e.g. Temperature), by computing new values at each point from the average of neighboring points.
  - Common, useful algorithm
  - Example: Solve Laplace equation in 2D: $\boldsymbol{\nabla^2 f(x,y) = 0}$



$$A_{k+1}(i,j) = \frac{A_k(i-1,j) + A_k(i+1,j) + A_k(i,j-1) + A_k(i,j+1)}{4}$$

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Jacobi Iteration C Code

```
while ( error > tol && iter < iter_max )
{
  error=0.0;

  for( int j = 1; j < n-1; j++) {
    for(int i = 1; i < m-1; i++) {


      Anew[j][i] = 0.25 * (A[j][i+1] + A[j][i-1] +
                           A[j-1][i] + A[j+1][i]);


      error = max(error, abs(Anew[j][i] - A[j][i]);
    }
  }

  for( int j = 1; j < n-1; j++) {
    for( int i = 1; i < m-1; i++ ) {
      A[j][i] = Anew[j][i];
    }
  }

  iter++;
}
```

- Iterate until converged
- Iterate across matrix elements
- Calculate new value from neighbors
- Compute max error for convergence
- Swap input/output arrays

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First Attempt: OpenACC C

```
while ( error > tol && iter < iter_max ) {
  error=0.0;

#pragma acc kernels
  for( int j = 1; j < n-1; j++) {
    for(int i = 1; i < m-1; i++) {

      Anew[j][i] = 0.25 * (A[j][i+1] + A[j][i-1] +
                           A[j-1][i] + A[j+1][i]);

      error = max(error, abs(Anew[j][i] - A[j][i]);
    }
  }

#pragma acc kernels
  for( int j = 1; j < n-1; j++) {
    for( int i = 1; i < m-1; i++ ) {
      A[j][i] = Anew[j][i];
    }
  }

  iter++;
}
```

Execute GPU kernel for loop nest

Execute GPU kernel for loop nest

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OpenMP C Code

```
while ( error > tol && iter < iter_max ) {
  error=0.0;

#pragma omp parallel for shared(m, n, Anew, A)
  for( int j = 1; j < n-1; j++) {
    for(int i = 1; i < m-1; i++) {

      Anew[j][i] = 0.25 * (A[j][i+1] + A[j][i-1] +
                           A[j-1][i] + A[j+1][i]);

      error = max(error, abs(Anew[j][i] - A[j][i]);
    }
  }

#pragma omp parallel for shared(m, n, Anew, A)
  for( int j = 1; j < n-1; j++) {
    for( int i = 1; i < m-1; i++ ) {
      A[j][i] = Anew[j][i];
    }
  }

  iter++;
}
```

Parallelize loop across CPU threads

Parallelize loop across CPU threads

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First Attempt: Compiler output (C)

```
pgcc  -acc -ta=nvidia -Minfo=accel -o laplace2d_acc laplace2d.c
main:
    57, Generating copyin(A[:4095][:4095])
        Generating copyout(Anew[1:4094][1:4094])
        Generating compute capability 1.3 binary
        Generating compute capability 2.0 binary
    58, Loop is parallelizable
    60, Loop is parallelizable
        Accelerator kernel generated
        58, #pragma acc loop worker, vector(16) /* blockIdx.y threadIdx.y */
        60, #pragma acc loop worker, vector(16) /* blockIdx.x threadIdx.x */
            Cached references to size [18x18] block of 'A'
            CC 1.3 : 17 registers; 2656 shared, 40 constant, 0 local memory bytes; 75% occupancy
            CC 2.0 : 18 registers; 2600 shared, 80 constant, 0 local memory bytes; 100% occupancy
        64, Max reduction generated for error
    69, Generating copyout(A[1:4094][1:4094])
        Generating copyin(Anew[1:4094][1:4094])
        Generating compute capability 1.3 binary
        Generating compute capability 2.0 binary
    70, Loop is parallelizable
    72, Loop is parallelizable
        Accelerator kernel generated
        70, #pragma acc loop worker, vector(16) /* blockIdx.y threadIdx.y */
        72, #pragma acc loop worker, vector(16) /* blockIdx.x threadIdx.x */
            CC 1.3 : 8 registers; 48 shared, 8 constant, 0 local memory bytes; 100% occupancy
            CC 2.0 : 10 registers; 8 shared, 56 constant, 0 local memory bytes; 100% occupancy
```

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First Attempt: Performance

CPU: Intel Xeon X5680
6 Cores @ 3.33GHz

GPU: NVIDIA Tesla M2070

Execution	Time (s)	Speedup
CPU 1 OpenMP thread	69.80	--
CPU 2 OpenMP threads	44.76	1.56x
CPU 4 OpenMP threads	39.59	1.76x
CPU 6 OpenMP threads	39.71	1.76x
OpenACC GPU	162.16	0.24x FAIL

Speedup vs. 1 CPU core

Speedup vs. 6 CPU cores

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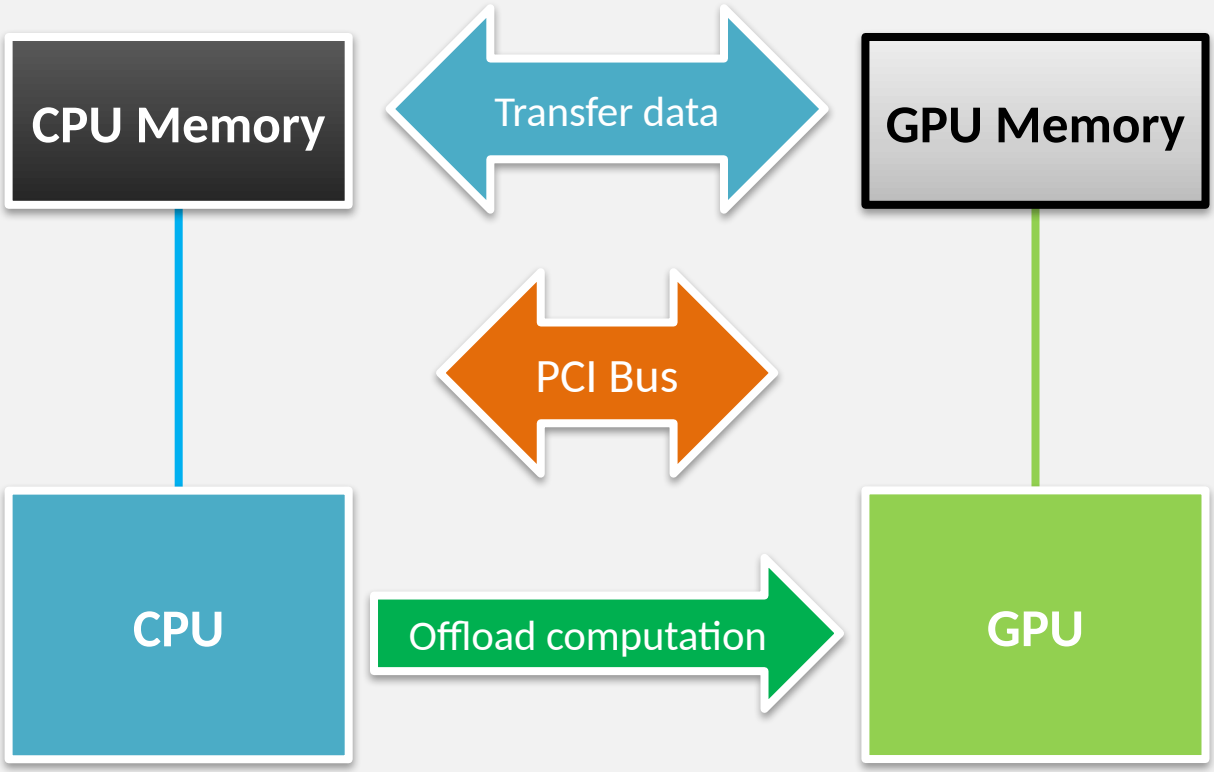
Basic Concepts



For efficiency, decouple data movement and compute off-load

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Excessive Data Transfers

```
while ( error > tol && iter < iter_max )
{
  error=0.0;
```

A, Anew resident on host

Copy

```
#pragma acc kernels
```

A, Anew resident on accelerator

```
  for( int j = 1; j < n-1; j++) {
    for( int i = 1; i < m-1; i++) {
      Anew[j][i] = 0.25 * (A[j][i+1] + A[j][i-1] +
                           A[j-1][i] + A[j+1][i]);
      error = max(error, abs(Anew[j][i] - A[j]
[i]);
    }
  }
```

These copies happen every iteration of the outer while loop!*

A, Anew resident on accelerator

Copy

A, Anew resident on host

```
    ...
}
```

*Note: there are two #pragma acc kernels, so there are 4 copies per while loop iteration!

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Data Management

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Data Construct

Fortran

```
!$acc data [clause …]
    structured block
!$acc end data
```

C

```
#pragma acc data [clause …]
    { structured block }
```

General Clauses

```
if( condition )
async( expression )
```

Manage data movement. Data regions may be nested.

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Data Clauses

`copy ( list )`	Allocates memory on GPU and copies data from host to GPU when entering region and copies data to the host when exiting region.
`copyin ( list )`	Allocates memory on GPU and copies data from host to GPU when entering region.
`copyout ( list )`	Allocates memory on GPU and copies data to the host when exiting region.
`create ( list )`	Allocates memory on GPU but does not copy.
`present ( list )`	Data is already present on GPU from another containing data region.

and `present_or_copy[in|out]`, `present_or_create`, `deviceptr`.

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Array Shaping

- Compiler sometimes cannot determine size of arrays
  - Must specify explicitly using data clauses and array "shape"

- C

```
#pragma acc data copyin(a[0:size-1]), copyout(b[s/4:3*s/4])
```

- Fortran

```
!$pragma acc data copyin(a(1:size)), copyout(b(s/4:3*s/4))
```

- Note: data clauses can be used on `data`, `kernels` or `parallel`

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Update Construct

Fortran

```
!$acc update [clause …]
```

C

```
#pragma acc update [clause …]
```

Clauses

```
host( list )
device( list )
if( expression )
async( expression )
```

Used to update existing data after it has changed in its corresponding copy (e.g. update device copy after host copy changes)

Move data from GPU to host, or host to GPU.
Data movement can be conditional, and asynchronous.

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Second Attempt: OpenACC C

```
#pragma acc data copy(A), create(Anew)
while ( error > tol && iter < iter_max ) {
  error=0.0;

#pragma acc kernels
  for( int j = 1; j < n-1; j++) {
    for(int i = 1; i < m-1; i++) {

      Anew[j][i] = 0.25 * (A[j][i+1] + A[j][i-1] +
                           A[j-1][i] + A[j+1][i]);

      error = max(error, abs(Anew[j][i] - A[j][i]);
    }
  }

#pragma acc kernels
  for( int j = 1; j < n-1; j++) {
    for( int i = 1; i < m-1; i++ ) {
      A[j][i] = Anew[j][i];
    }
  }

  iter++;
}
```

Copy A in at beginning of loop, out at end. Allocate Anew on accelerator

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Second Attempt: Performance

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6 Cores @ 3.33GHz

GPU: NVIDIA Tesla M2070

Execution	Time (s)	Speedup
CPU 1 OpenMP thread	69.80	--
CPU 2 OpenMP threads	44.76	1.56x
CPU 4 OpenMP threads	39.59	1.76x
CPU 6 OpenMP threads	39.71	1.76x
OpenACC GPU	13.65	2.9x

Speedup vs. 1 CPU core

Speedup vs. 6 CPU cores

Note: same code runs in 9.78s on NVIDIA Tesla M2090 GPU

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Further speedups

- OpenACC gives us more detailed control over parallelization
  - Via gang, worker, and vector clauses
- By understanding more about OpenACC execution model and GPU hardware organization, we can get higher speedups on this code
- By understanding bottlenecks in the code via profiling, we can reorganize the code for higher performance
- Will tackle these in later exercises

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Finding Parallelism in your code

- (Nested) for loops are best for parallelization
- Large loop counts needed to offset GPU/memcpy overhead
- Iterations of loops must be <u>independent</u> of each other
  - To help compiler: `restrict` keyword (C), independent clause
- Compiler must be able to figure out sizes of data regions
  - Can use directives to explicitly control sizes
- Pointer arithmetic should be avoided if possible
  - Use subscripted arrays, rather than pointer-indexed arrays.
- Function calls within accelerated region must be inlineable.

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Tips and Tricks

- (PGI) Use time option to learn where time is being spent
  `-ta=nvidia,time`
- Eliminate pointer arithmetic
- Inline function calls in directives regions
  (PGI): `-inline` or `-inline,levels(<N>)`
- Use contiguous memory for multi-dimensional arrays
- Use data regions to avoid excessive memory transfers
- Conditional compilation with `_OPENACC` macro

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OpenACC Learning Resources

- OpenACC info, specification, FAQ, samples, and more
  - http://openacc.org
- PGI OpenACC resources
  - http://www.pgroup.com/resources/accel.htm

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OpenACC API

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Kernels Construct

Fortran

```
!$acc kernels [clause …]
    structured block
!$acc end kernels
```

C

```
#pragma acc kernels [clause …]
    { structured block }
```

Clauses

```
if( condition )
async( expression )
```

Also any data clause

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Kernels Construct

Each loop executed as a separate kernel on the GPU.

```
!$acc kernels
     do i=1,n
        a(i) = 0.0
        b(i) = 1.0
        c(i) = 2.0
     end do

     do i=1,n
        a(i) = b(i) + c(i)
     end do
!$acc end kernels
```

kernel 1

kernel 2

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Parallel Construct

Fortran

```
!$acc parallel [clause …]
    structured block
!$acc end parallel
```

C

```
#pragma acc parallel [clause …]
    { structured block }
```

Clauses

```
if( condition )
async( expression )
num_gangs( expression )
num_workers( expression )
vector_length( expression )
```

```
private( list )
firstprivate( list )
reduction( operator:list )
```

Also any data clause

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Parallel Clauses

`num_gangs ( expression )`	Controls how many parallel gangs are created (CUDA `gridDim`).
`num_workers ( expression )`	Controls how many workers are created in each gang (CUDA `blockDim`).
`vector_length ( list )`	Controls vector length of each worker (SIMD execution).
`private( list )`	A copy of each variable in list is allocated to each gang.
`firstprivate ( list )`	`private` variables initialized from host.
`reduction( operator:list )`	`private` variables combined across gangs.

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Loop Construct

Fortran

```
!$acc loop [clause …]
    loop
!$acc end loop
```

C

```
#pragma acc loop [clause …]
    { loop }
```

Combined directives

```
!$acc parallel loop [clause …]
!$acc kernels loop [clause …]
```

```
!$acc parallel loop [clause …]
!$acc kernels loop [clause …]
```

Detailed control of the parallel execution of the following loop.

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Loop Clauses

`collapse( n )`	Applies directive to the following `n` nested loops.
`seq`	Executes the loop sequentially on the GPU.
`private( list )`	A copy of each variable in list is created for each iteration of the loop.
`reduction( operator:list )`	`private` variables combined across iterations.

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Loop Clauses Inside parallel Region

`gang`	Shares iterations across the gangs of the parallel region.
`worker`	Shares iterations across the workers of the gang.
`vector`	Execute the iterations in SIMD mode.

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Loop Clauses Inside kernels Region

`gang [( num_gangs )]`	Shares iterations across across at most *num_gangs* gangs.
`worker [( num_workers )]`	Shares iterations across at most *num_workers* of a single gang.
`vector [( vector_length )]`	Execute the iterations in SIMD mode with maximum *vector_length*.
`independent`	Specify that the loop iterations are independent.

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Other Directives

`cache` construct	Cache data in software managed data cache (CUDA shared memory).
`host_data` construct	Makes the address of device data available on the host.
`wait` directive	Waits for asynchronous GPU activity to complete.
`declare` directive	Specify that data is to allocated in device memory for the duration of an implicit data region created during the execution of a subprogram.

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Runtime Library Routines

Fortran

```
use openacc
#include "openacc_lib.h"

acc_get_num_devices
acc_set_device_type
acc_get_device_type
acc_set_device_num
acc_get_device_num
acc_async_test
acc_async_test_all
```

C

```
#include "openacc.h"

acc_async_wait
acc_async_wait_all
acc_shutdown
acc_on_device
acc_malloc
acc_free
```

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Environment and Conditional Compilation

`ACC_DEVICE` *`device`*	Specifies which device type to connect to.
`ACC_DEVICE_NUM` `num`	Specifies which device number to connect to.
`_OPENACC`	Preprocessor directive for conditional compilation. Set to OpenACC version

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